Automated Student Performance Prediction Using AutoML

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***Abstract*—Student performance prediction plays an important role in identifying students who may require early academic support. Traditional machine learning approaches often require manual selection of algorithms, features, and hyperparameters. This paper proposes an Automated Machine Learning (AutoML) framework for predicting student academic performance. The proposed system performs data preprocessing, feature selection, model comparison, and hyperparameter optimization to automatically identify a suitable predictive model. Student academic and behavioral attributes are used as input for prediction. The performance of the selected model is evaluated using appropriate machine learning metrics. The proposed framework aims to reduce manual effort in model development and provide an efficient approach for student performance prediction. The system can support educators and institutions in identifying students who may require additional academic assistance.**

***Index Terms*—Student Performance Prediction, AutoML, Machine Learning, Educational Data Mining, Academic Performance**

I. Introduction

Student academic performance is an important indicator of learning outcomes. Predicting student performance at an early stage can help educational institutions identify students who may require additional academic support.

The increasing availability of student data from learning management systems, academic records, and assessment platforms provides an opportunity to develop data-driven prediction systems. However, the effectiveness of prediction depends on selecting suitable features and machine learning models for the given dataset. Manual experimentation with different algorithms and their parameters can be time-consuming.

Automated Machine Learning (AutoML) can reduce this manual effort by automating important stages of the machine learning workflow. These stages may include data preprocessing, feature selection, model selection, and hyperparameter optimization. By evaluating multiple candidate models automatically, AutoML can help identify a suitable model for a particular student dataset.

The objective of this work is to develop an automated framework for predicting student academic performance using AutoML. The proposed framework analyzes relevant student attributes, applies preprocessing and feature selection, evaluates multiple machine learning models, and selects a suitable model based on its predictive performance. The approach aims to reduce manual model development effort and provide an efficient method for identifying students who may require academic support.

II. Related Work

Umamaheswari et al. proposed a machine learning approach for predicting student success using a modified Support Vector Machine (SVM) model [1]. The study focuses on student result prediction as a classification problem and evaluates the proposed approach using accuracy, recall, and F1-score. The study reported 84% accuracy, 95% recall, and 82% F1-score. The findings demonstrate the potential of machine learning techniques for predicting student academic outcomes.

Paddalwar et al. proposed a machine learning approach for predicting students' academic grades using a hybrid feature selection method [2]. The study uses student data collected from academic records, classroom observations, and learning management systems. Multiple feature selection techniques are combined with machine learning algorithms to identify relevant factors for academic performance prediction. The study highlights the importance of selecting appropriate features for improving student grade prediction.

Franciskovic et al. investigated the prediction of students' final examination scores using their engagement with preclass activities in a flipped classroom environment [3]. The study applies learning analytics and machine learning techniques to analyze students' learning traces and engagement patterns. The findings indicate that student engagement data can provide useful information for predicting final examination

performance. This study demonstrates the importance of behavioral and learning-activity features in student performance prediction.

Alrasheed et al. investigated the relationship between English proficiency and academic performance among software engineering students [4]. The study analyzes English proficiency as a factor that may influence students' academic outcomes. The findings indicate a relationship between English proficiency and academic performance, suggesting that non-academic attributes can also provide useful information when analyzing student performance. This work supports the inclusion of diverse student-related features in performance prediction systems.

Kaur and Singh analyzed the relationship between students' study habits and their academic performance [5]. The study compares different study-related behaviors and examines their association with academic outcomes. The findings suggest that study habits can be an important factor influencing student performance. This work supports the use of behavioral attributes as additional features in student performance prediction systems.

III. Proposed Methodology

The proposed system aims to automate the process of predicting student academic performance using an Automated Machine Learning (AutoML) framework. The system takes relevant student academic and behavioral attributes as input and processes them through multiple stages, including data preprocessing, feature selection, model selection, and hyperparameter optimization. The overall workflow is designed to reduce manual effort in selecting and tuning machine learning models.

The proposed methodology consists of five major stages: data collection, data preprocessing, feature selection, automated model selection and optimization, and performance evaluation. The student dataset is first collected from a suitable educational dataset containing relevant attributes related to academic performance. The collected data is then preprocessed to handle missing values, categorical variables, and inconsistent data.

After preprocessing, relevant features are selected to reduce unnecessary information and improve the predictive capability of the models. The processed dataset is then provided to the AutoML framework, which evaluates multiple machine learning algorithms and automatically performs hyperparameter optimization. The models are compared based on appropriate evaluation metrics, and the best-performing model is selected for student performance prediction.

The final selected model is used to predict the academic performance of students. The prediction results can help educators identify students who may require additional academic support and enable early intervention.

A. Dataset and Data Preprocessing

The student performance dataset contains academic, demographic, and behavioral attributes that can be used to analyze and predict student academic outcomes. The relevant input features are selected based on their relationship with student performance.

Before applying machine learning models, the dataset is preprocessed to improve data quality and ensure that the features are suitable for model training. Missing values are handled appropriately, categorical attributes are converted into numerical representations, and unnecessary or redundant features are removed. Numerical features can also be normalized or standardized when required by the selected machine learning algorithms.

After preprocessing, the dataset is divided into training and testing sets. The training data is used by the AutoML framework to identify suitable machine learning models, while the testing data is reserved for evaluating the final selected model.

B. AutoML Model Selection and Optimization

The AutoML framework is used to automate the selection and optimization of machine learning models for student performance prediction. Instead of manually selecting a single algorithm, the framework evaluates multiple candidate models using the training dataset.

The AutoML process performs model training and hyperparameter optimization to identify an appropriate configuration for the given dataset. Different models are compared using suitable evaluation metrics, and the configuration providing the best predictive performance is selected. This automated approach reduces the time and manual effort required for model experimentation and tuning.

The selected model is then trained using the available training data and used to generate predictions on unseen student data. This enables the proposed system to provide an automated and efficient approach for student performance prediction.

C. Performance Evaluation

The performance of the selected model is evaluated using appropriate classification or regression metrics, depending on the target variable used for student performance prediction. For classification-based prediction, metrics such as accuracy, precision, recall, and F1-score can be used to measure the predictive performance of the model. For regression-based prediction, metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) can be considered.

The evaluation results are used to determine the effectiveness of the automatically selected model on unseen student data. The model with suitable predictive performance is considered for the final student performance prediction system.

IV. Results and Discussion

The proposed AutoML framework is designed to evaluate multiple machine learning models for student performance prediction and identify a suitable model based on its predictive performance. The evaluation considers the performance of the

candidate models on the test dataset using appropriate machine learning metrics.

The preprocessing stage prepares the student dataset for model training by handling missing values, converting categorical attributes into numerical representations, and selecting relevant features. These steps help provide a consistent input for the machine learning models and reduce the effect of irrelevant or redundant information.

The AutoML framework automatically compares different machine learning algorithms and performs hyperparameter optimization. This reduces the requirement for manually selecting an algorithm and tuning its parameters. The selected model is then evaluated on unseen test data to determine its ability to generalize to new student records.

The final experimental results will be presented using evaluation metrics such as accuracy, precision, recall, and F1-score for classification-based prediction. If the target variable is treated as a continuous value, regression metrics such as MAE, MSE, and RMSE will be used. The comparison of these metrics will help determine the effectiveness of the selected model for student performance prediction.

The results can also be analyzed to identify the student attributes that contribute to prediction performance. Such analysis can provide useful insights into factors associated with academic outcomes and support the development of early academic intervention strategies.

V. Conclusion

This paper presented an Automated Machine Learning (AutoML) framework for predicting student academic performance. The proposed approach automates important stages of the machine learning workflow, including data preprocessing, feature selection, model selection, and hyperparameter optimization. By reducing the need for manual model experimentation, the framework provides an efficient approach to developing student performance prediction systems.

The proposed system can help educational institutions identify students who may require additional academic support at an early stage. Future work can focus on evaluating the framework using larger and more diverse educational datasets, incorporating additional student-related features, and improving the interpretability of the prediction results.

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